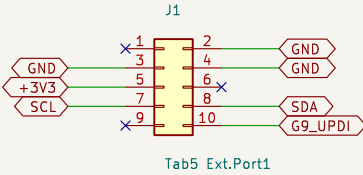
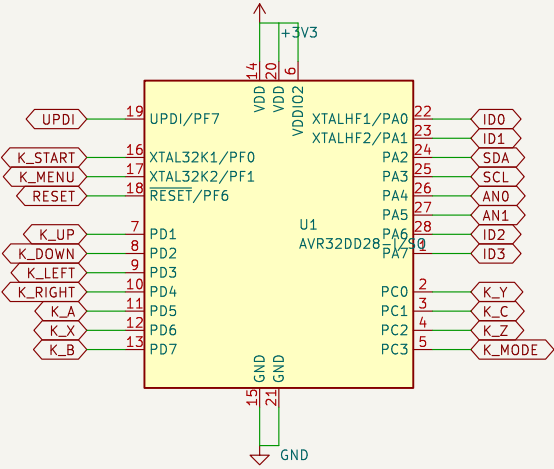


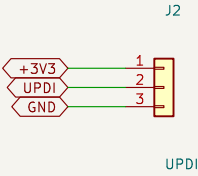
MCU – AVR32DD28 (SOIC–28) running firmware\_avrdd (I2C client 0x6D, RetroPad Ext. Port 1) – verify pin 1 orientation before fab

Buttons – one MCU pin each, switch to GND (internal pull–up); slot table in firmware\_avrdd/pinmap.h

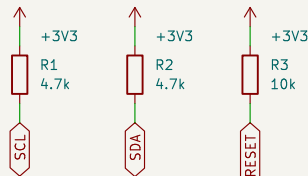
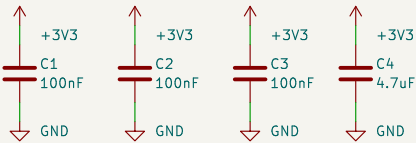
PD1–PD7, PC0–PC3, PF0–PF1: one button per pin (internal pull–ups)



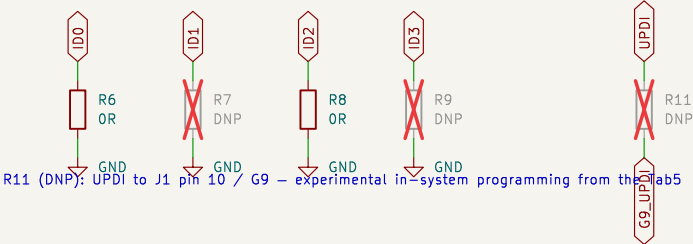
UPDI programming (SerialUPDI / MPLAB Snap)



Decoupling (VDD x2, VDDIO2), I2C pull–ups, RESET pull–up

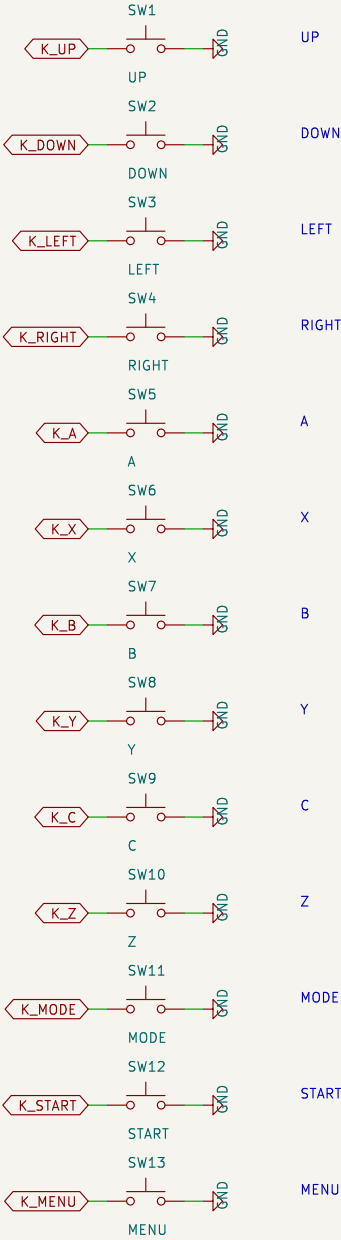
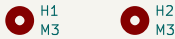


Console ID straps: OR to GND = bit set. Genesis 6–button (AVR DD) = ID 5 (fit R6, R8; R7, R9 DNP)



R11 (DNP): UPDI to J1 pin 10 / G9 – experimental in–system programming from the Tab5

Mounting holes (M3, 96 mm apart)



Tab5 keyboard–port board: STM32F030 + RetroPad firmware, I2C 0x6D  
Generated by kicad\_gen/gen\_sch.py from gen\_pcb.py and layout.py

**RetroESP32–P4**

Sheet: /  
File: retropad\_genesis\_dd.kicad\_sch

**Title: RetroPad Genesis 6–button (AVR DD) controller (console ID 5)**

Size: A3 Date: 2026–10–05

Rev: 0.1

KiCad E.D.A. kicad–cli 7.0.11+dfsg–1build4

Id: 1/1